

CH365 Chemical Engineering Thermodynamics

Lesson 16 Sensible Heat Effects

Ethylene Glycol Process

Slide 2

- Sensible heat effects are associated with temperature change
- Latent heat (phase changes) – no temperature change
- Heat of Reaction
- Heat of Mixing

“Sensible” Heat Effects

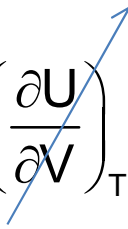
Sensible = No phase transitions, no chemical reactions, and no change in composition.

$$U = U(T, V)$$

Total differential introduced
in L10, Slide 8, page 138

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

0 for constant volume process,
ideal gases, or incompressible fluids



Mean Heat Capacity

Integral evaluated forms – “user-defined functions” – simplifies working with mixtures

These forms are used in later derivations in the textbook.

Ideal Gas Heat Capacity in Simulators

Improved function – used in professional simulators such as CC and Aspen+

F. A. Aly and L. L. Lee, "Self-Consistent Equations for calculating the Ideal Gas Heat Capacity, Enthalpy, and Entropy," *Fluid Phase Equilibria*, 1981, Vol. 6, Issues 3-4, pp. 169-179.

a, b, c, d & e are constants published and maintained by DIPPr (link on course web site).

$$C_p = C_p(T) = a + b \cdot \left(\frac{c/T}{\text{Sinh}[c/T]} \right)^2 + d \cdot \left(\frac{e/T}{\text{Cosh}[e/T]} \right)^2$$

hyperbolic sine hyperbolic cosine

DIPPr Eq 107

The screenshot illustrates the steps to find the DIPPr equation 107 for ideal gas heat capacity in Aspen+:

- Component List:** A small window showing "1 (62) Water". A blue arrow points to the "View/Edit" button.
- View/Edit Component Data:** A larger window with a sidebar. A blue arrow points to the "Heat Capacity Data" option.
- Library Heat Capacity Data:** A window showing data for "Water" (Component ID: 62).
 - Ideal Gas Heat Capacity (J/kmol-K):** The "Equation No." field is circled in red and contains the value "107".
 - Liquid Heat Capacity (J/kmol-K):** The "Equation No." field contains the value "100".
 - Solid Heat Capacity (J/kmol-K):** The "Equation No." field contains the value "100".
 - Coefficients:** A table of coefficients A through G is circled in red. For the ideal gas equation, the values are: A=33359, B=26798, C=2609.3, D=8888, E=1167.6, F=, G=.

CC/DIPPr eq 107 found here

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$$C_p = C_p(T) = a + b \cdot \left(\frac{c/T}{\sinh[c/T]} \right)^2 + d \cdot \left(\frac{e/T}{\cosh[e/T]} \right)^2$$

hyperbolic sine
hyperbolic cosine

DIPPr Eq 107

The screenshot shows the Aspen+ software interface. The 'Results - Pure Components' window is open, displaying a table of heat capacity values for Methane, Ethane, and Propane. A red circle highlights the values for Element 1, 2, 3, 4, and 5 for Methane, which correspond to the constants a, b, c, d, and e in the equation.

Component	METHANE	ETHANE	PROPANE
Element 1	33298	44256	59474
Element 2	79933	84737	126610
Element 3	2086.9	872.24	844.31
Element 4	41602	67130	86165
Element 5	991.96	2430.4	2482.7
Element 6	50	298.15	298.15
Element 7	1500	1500	1500

Aspen+/DIPPr eq 107 found here

Questions?